

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 21: Trigonometrical Identities

Previous year board questions, topic-wise: 13 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 The three basic identities

1. Assertion (A): If $\sin^2 A + \sin A = 1$, then $\cos^4 A + \cos^2 A = 1$. Reason (R): $1 - \sin^2 A = \cos^2 A$. (a) A true, R false (b) A false, R true (c) both true, and R is the correct reason for A (d) both true, but R is not the correct reason for A **ICSE 2025, 1 mark**
2. Statement 1: $\sin^2 \theta + \cos^2 \theta = 1$. Statement 2: $\operatorname{cosec}^2 \theta + \cot^2 \theta = 1$. Which is valid? (a) only 1 (b) only 2 (c) both (d) neither **ICSE 2024, 1 mark**
3. $(1 + \sin A)(1 - \sin A)$ equals: (a) $\operatorname{cosec}^2 A$ (b) $\sin^2 A$ (c) $\sec^2 A$ (d) $\cos^2 A$ **ICSE 2023, 1 mark**

2 Proving identities

4. Prove that $\frac{(\cot A + \tan A - 1)(\sin A + \cos A)}{\sin^3 A + \cos^3 A} = \sec A \operatorname{cosec} A$. **ICSE 2025, 4 marks**
5. Factorise $\sin^3 \theta + \cos^3 \theta$. Hence prove that $\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} + \sin \theta \cos \theta = 1$. **ICSE 2024, 4 marks**
6. Prove that $(\sin^2 \theta - 1)(\tan^2 \theta + 1) + 1 = 0$. **ICSE 2023, 4 marks**
7. Prove that $\frac{\sin A}{1 + \cot A} - \frac{\cos A}{1 + \tan A} = \sin A - \cos A$. **ICSE 2020, 3 marks**
8. Prove that $\left(\frac{1 - \tan \theta}{1 - \cot \theta}\right)^2 = \tan^2 \theta$. **ICSE 2020, 4 marks**
9. Prove that $(\operatorname{cosec} \theta - \sin \theta)(\sec \theta - \cos \theta)(\tan \theta + \cot \theta) = 1$. **ICSE 2019, 3 marks**
10. Prove that $\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta} = \tan \theta + \cot \theta$. **ICSE 2018, 3 marks**
11. Prove that $(1 + \cot \theta - \operatorname{cosec} \theta)(1 + \tan \theta + \sec \theta) = 2$. **ICSE 2018, 4 marks**
12. Prove that $\frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta} = \tan \theta$. **ICSE 2017, 3 marks**

3 Complementary angles

13. Without using tables, evaluate $\sin^2 28^\circ + \sin^2 62^\circ + \tan^2 38^\circ - \cot^2 52^\circ + \frac{1}{4} \sec^2 30^\circ$. **ICSE 2017, 3 marks**

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